

Don't Just Listen, Try Planning

Graph-based Retrieval-Generation Agent for Long-form Audio Meeting Understanding

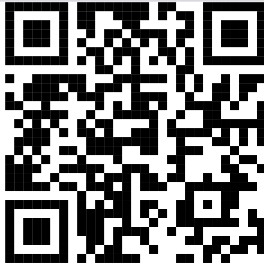
Quanwei Tang, Dong Zhang*, Shoushan Li, Guodong Zhou

School of Computer Science & Technology, NLP Lab, Soochow University Jiangsu Key Lab of Language Computing

[LongAudioQA Dataset](#) [Multi-dimensional Graph](#) [Agent Planning](#) [Acoustic-aware QA](#)

Code & Data

Scan for GRGA resources and LongAudioQA.

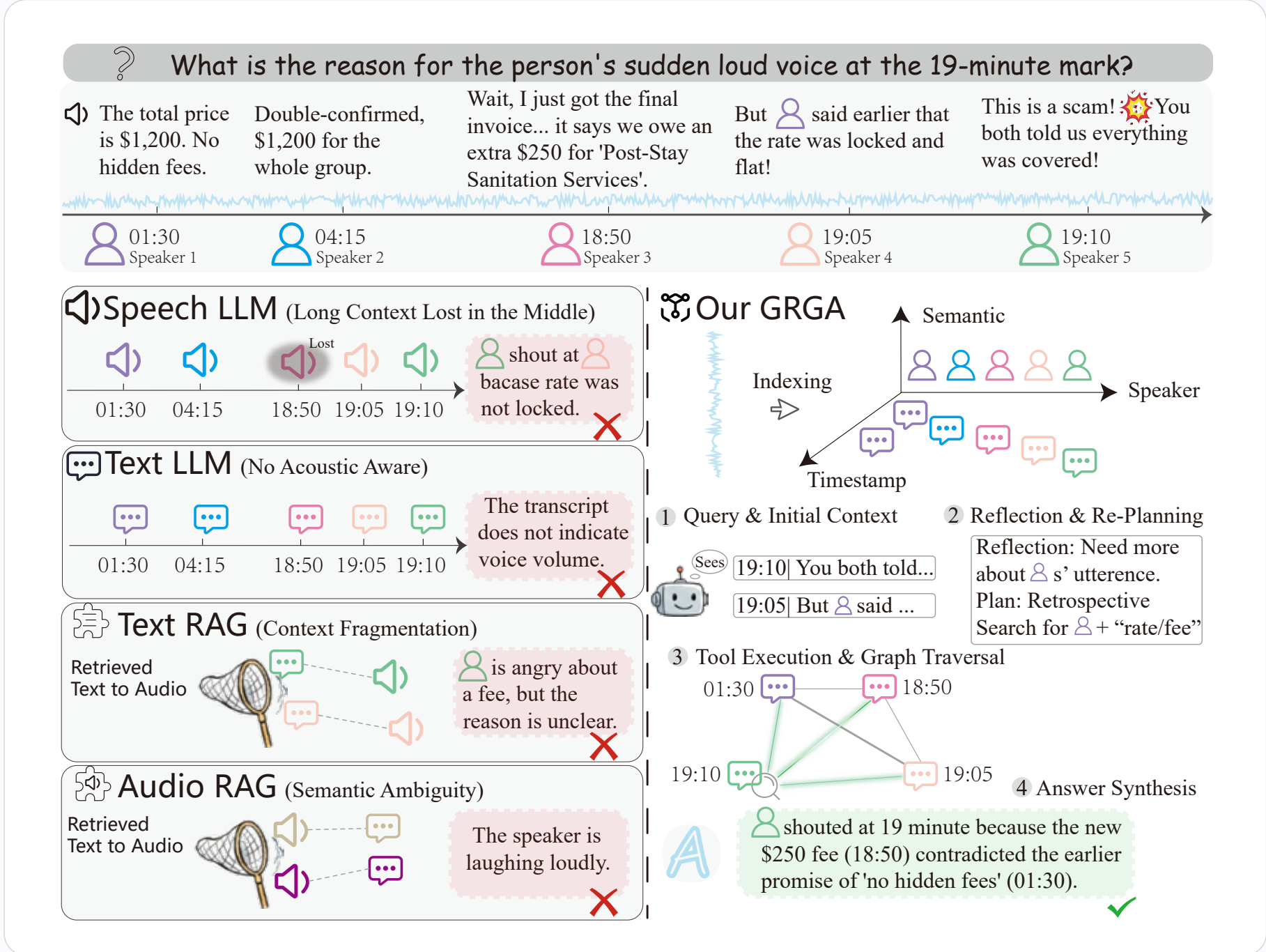


Takeaway:

Plan over semantic, speaker, timestamp, and acoustic evidence instead of retrieving one flat chunk.

Why Long Audio QA Breaks

- Long-form audio meeting understanding lacks dedicated QA benchmarks.
- Speech LLMs map audio toward text, losing paralinguistic cues such as loudness, emotion, or speaker tension.
- Flat RAG retrieves local transcript chunks, but meeting answers often require long-range evidence and speaker/time links.

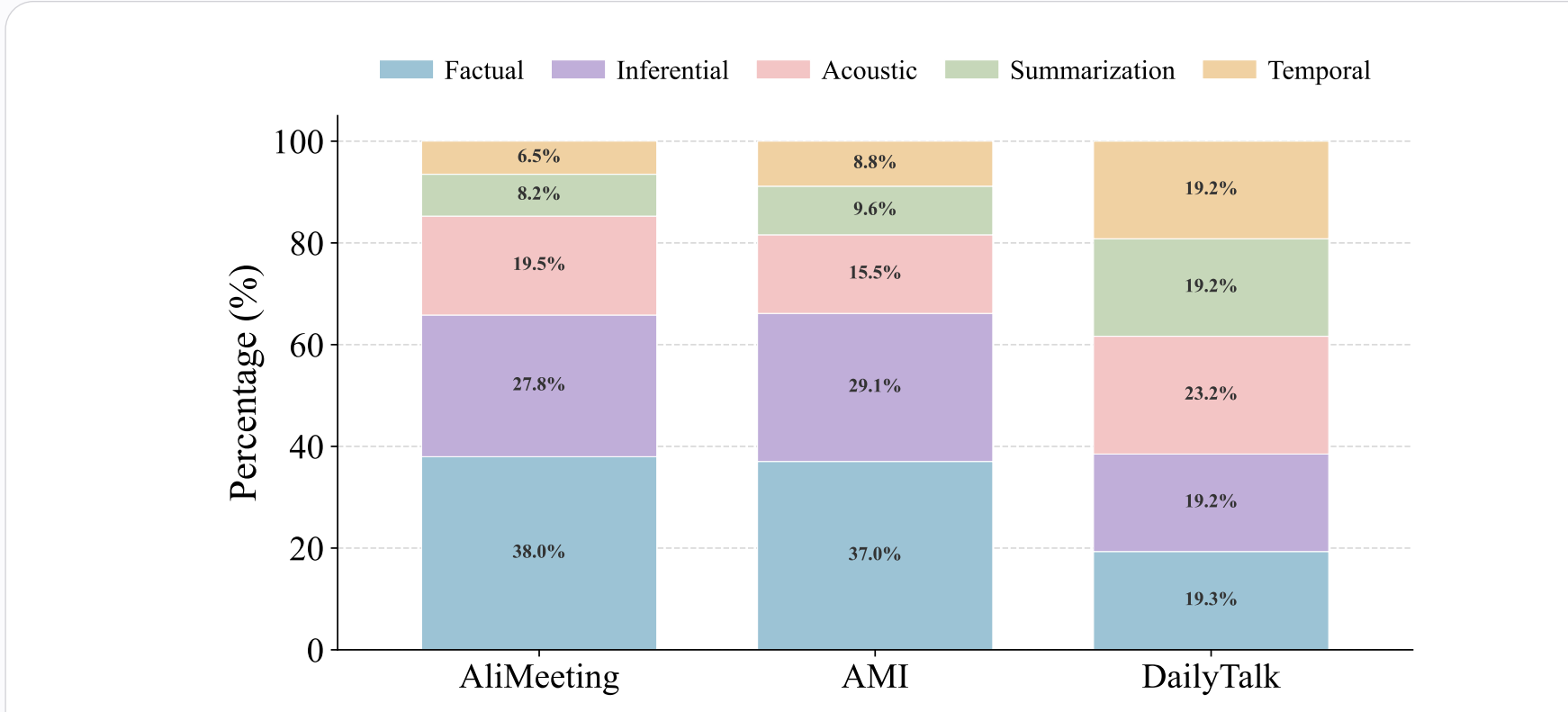


GRGA indexes conversation along semantic, speaker, timestamp, and acoustic dimensions, enabling grounded reasoning over dispersed evidence.

LongAudioQA

- New QA dataset for long-form meeting and conversation understanding.
- Built from AliMeeting, AMI Meeting, and DailyTalk with human verification.
- Five skills: factual retrieval, inferential reasoning, temporal grounding, summarization, and acoustic-aware reasoning.

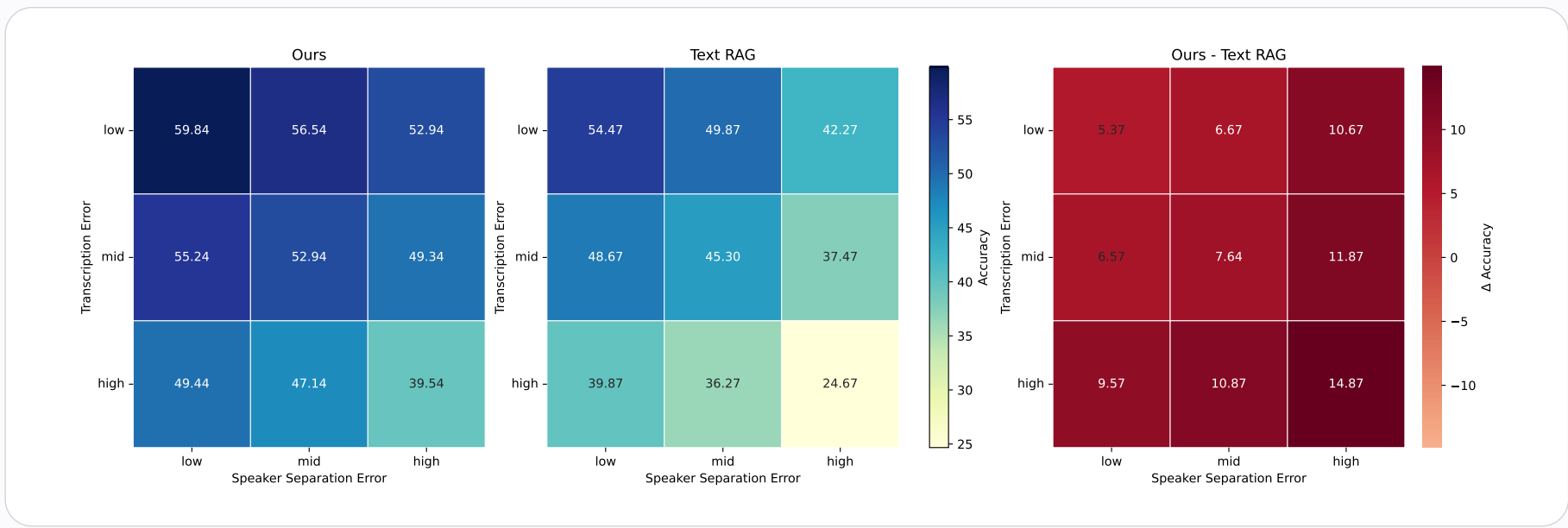
Dataset	Dur(h)	Avg(s)	Turns	Q&A
AliMeeting	14.91	1,935	815	3,013
AMI	18.24	1,930	757	3,243
DailyTalk	21.59	610	186	10,200



Question Taxonomy

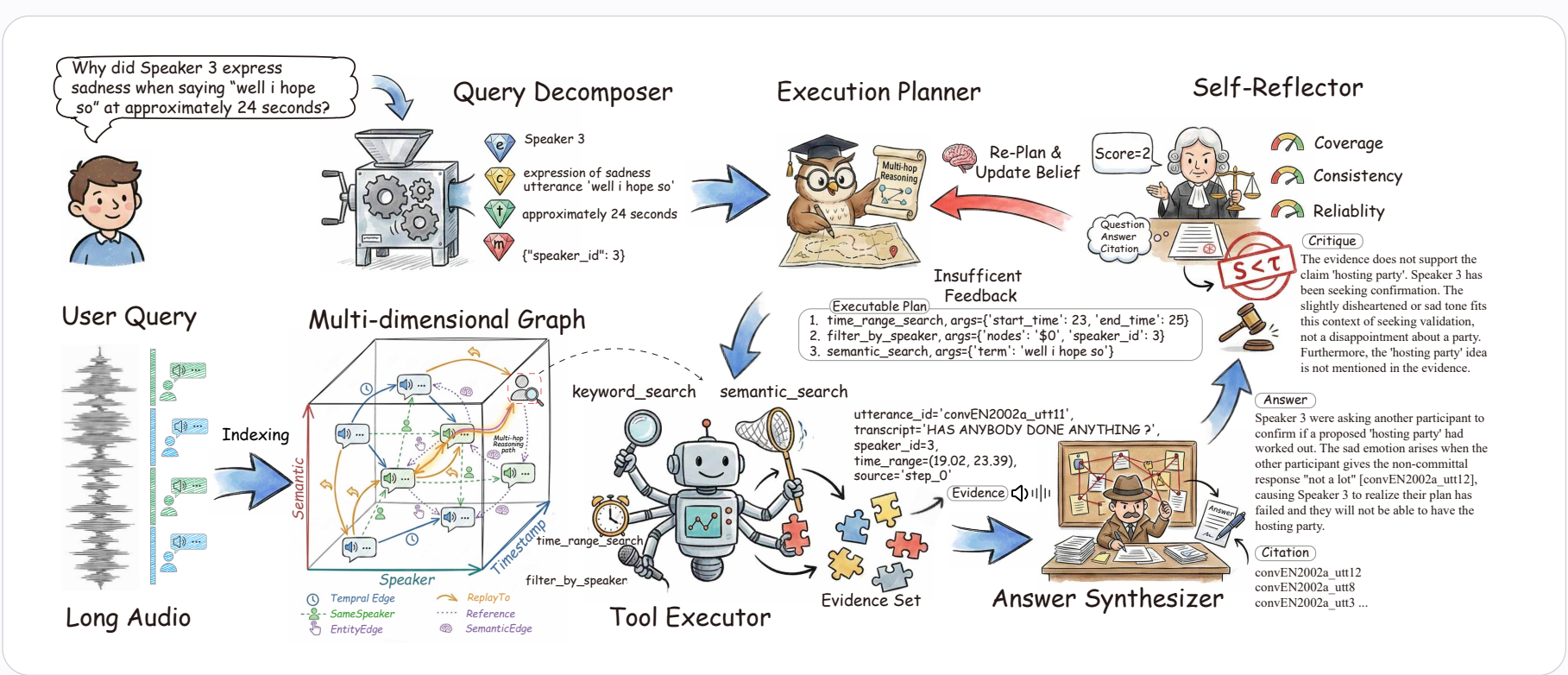
Factual	Entity retrieval, attribute association, keyword locating.
Inferential	Multi-hop links between early problems, later discussions, and final decisions.
Temporal	Absolute localization, relative sequencing, frequency statistics in time windows.
Summarization	Topic consensus and speaker contribution aggregation.
Acoustic-aware	Align emotion, intensity, laughter, or volume spikes with semantic context.

Robustness Under Noise



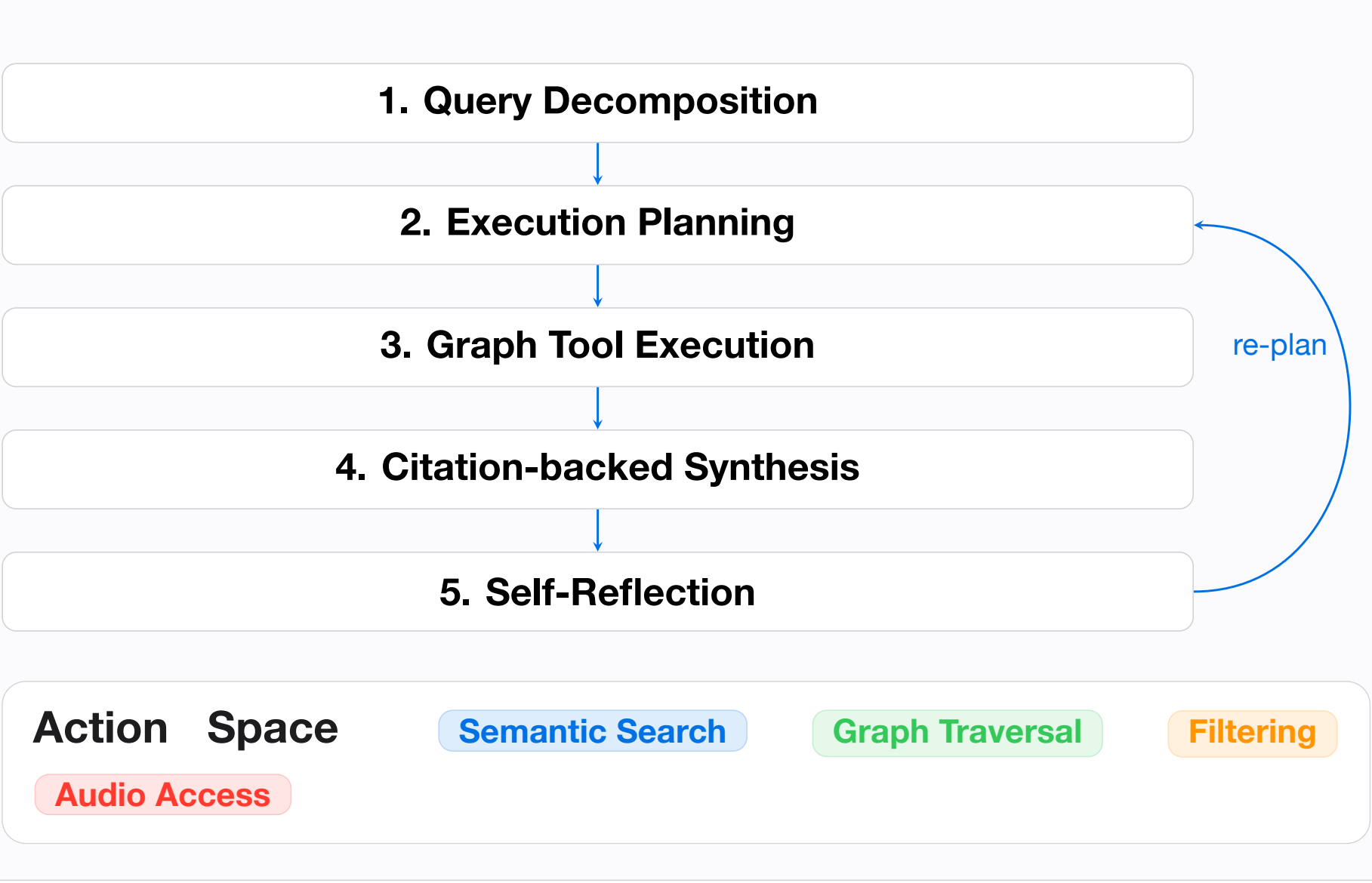
- Text RAG degrades sharply as ASR and diarization noise increase.
- GRGA acts as a structured noise buffer by validating evidence through graph links and reflection.

GRGA: Graph Retrieval + Planning

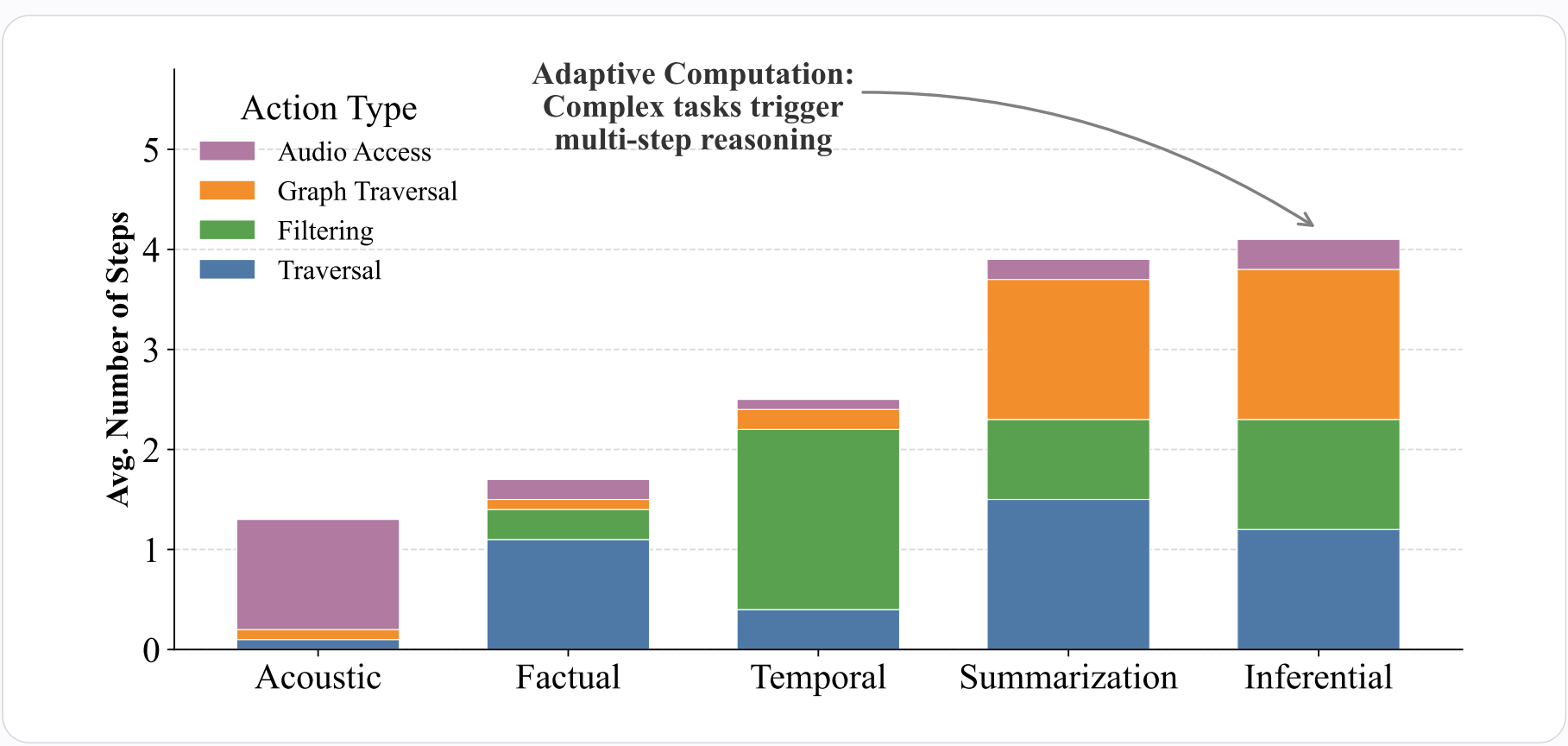


- Audio is converted into utterance nodes with text, speaker, timestamps, and audio references.
- Edges encode temporal adjacency, reply-to relations, same-speaker links, entity co-occurrence, and semantic coreference.
- The agent decomposes each query, plans tool calls, executes graph retrieval, synthesizes an answer, then reflects for support.

Reasoning Loop



Why Planning Helps



- Factual queries can terminate quickly with semantic search.
- Inferential and temporal questions trigger deeper graph traversal.
- Acoustic-aware questions selectively access raw audio when transcripts are insufficient.

Main Results

+40.7
AMI Inferential
vs Text RAG

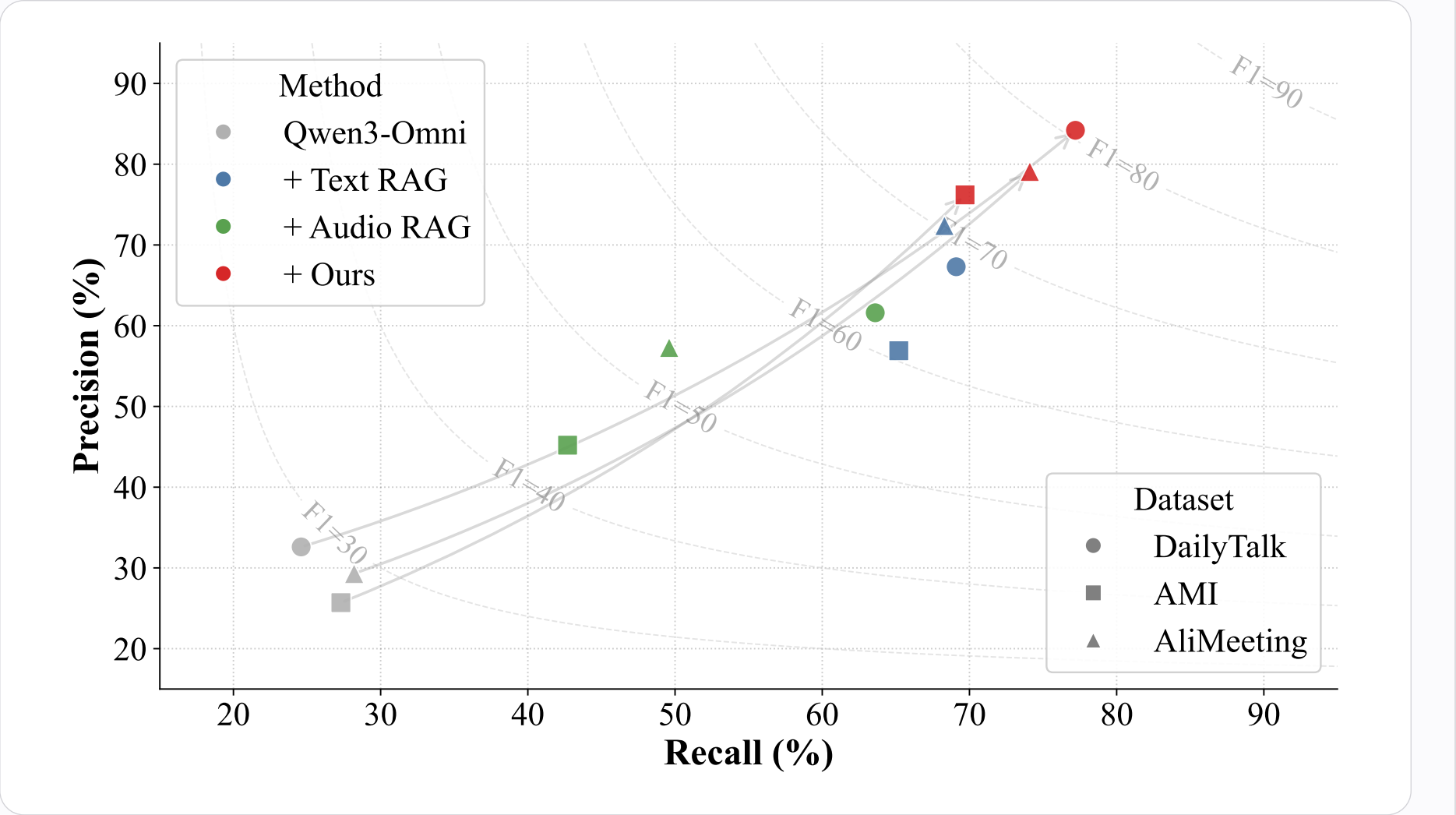
52.32
DailyTalk Acoustic
accuracy

61.10
DailyTalk Summarization
accuracy

AMI Method	Fact.	Infer.	Temp.	Summ.	Acou.
Qwen3-Omni speech	20.96	26.38	15.12	16.67	12.15
MiMo-Audio text	53.62	44.10	37.14	41.49	17.94
BGE-M3 Text RAG	32.56	24.56	23.78	25.36	25.34
CLASP Audio RAG	24.12	20.31	22.76	11.60	14.79
GRGA	59.46	65.29	38.56	48.46	35.68

Semantic Accuracy (%) with LLM-as-a-judge. GRGA is best across all AMI question types.

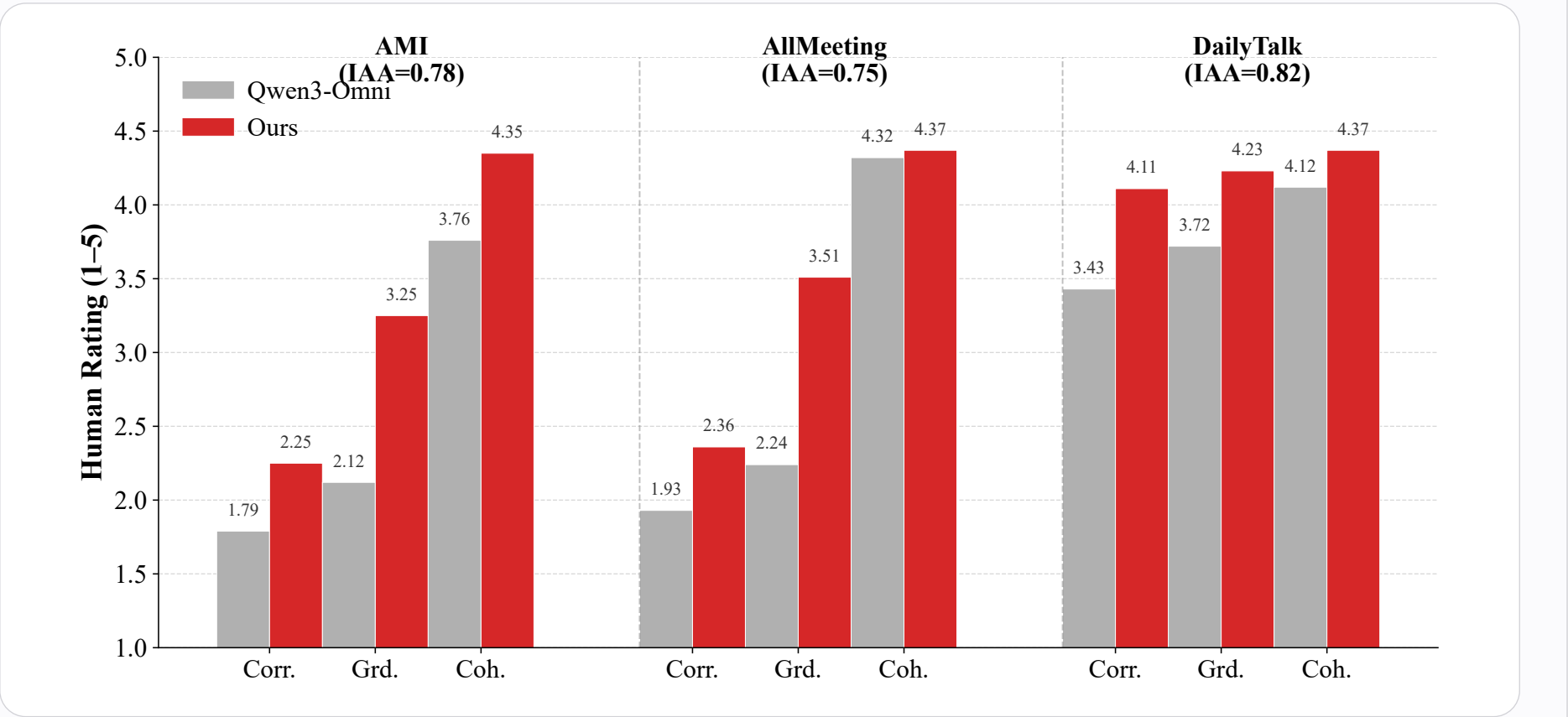
Grounded Answers



Method on AMI	Precision	Recall
Qwen3-Omni	25.7	27.3
+ Text RAG	56.9	65.2
+ Audio RAG	45.2	42.7
+ GRGA	76.2	69.7

Citation accuracy uses a $\pm 2s$ tolerance over timestamp evidence.

Human Evaluation



+0.97
Overall groundedness gain

+0.52
Overall correctness gain

Ablation Highlights

Removed Component	Avg.	Drop
Full GRGA	49.49	–
w/o Reflection	39.64	-9.84
w/o Graph Traversal	38.21	-11.27
w/o Audio Access	34.71	-14.78
w/o Semantic Search	16.28	-33.21

- Semantic search finds entry evidence; graph traversal connects dispersed clues.
- Audio access is crucial for acoustic-aware questions.
- Reflection prevents accepting unsupported first-pass answers.

Conclusion

Long-form audio QA needs planning, not just listening.

- Dataset:** LongAudioQA introduces long-context, multi-speaker, temporal, and acoustic-aware QA.
- Method:** GRGA turns meetings into multi-dimensional graphs and solves QA through tool-using agent planning.
- Evidence:** Improvements appear in accuracy, citation precision/recall, human groundedness, and noisy settings.

Future Directions Reduce iterative latency, strengthen upstream robustness, and extend beyond structured meetings to broader long-form audio.